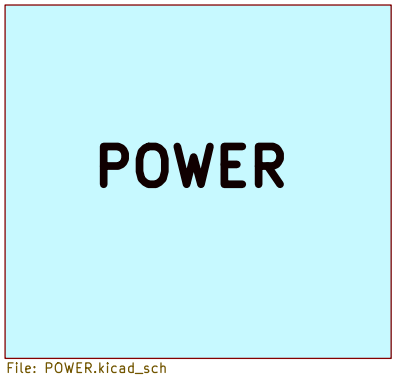
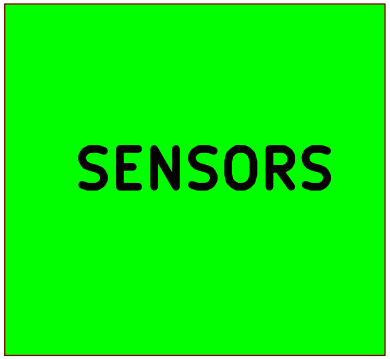


File: POWER.kicad_sch

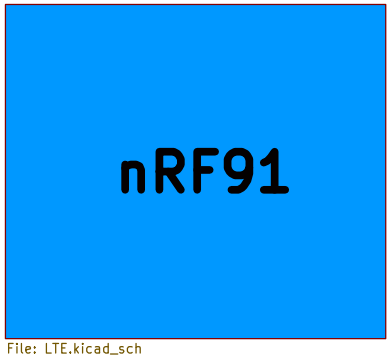


File: CPU.kicad_sch



File: SENSOR.kicad_sch

LTE



File: LTE.kicad_sch

amb82_conn



File: amb82_conn.kicad_sch

Carrier Board for AMB82 Mini

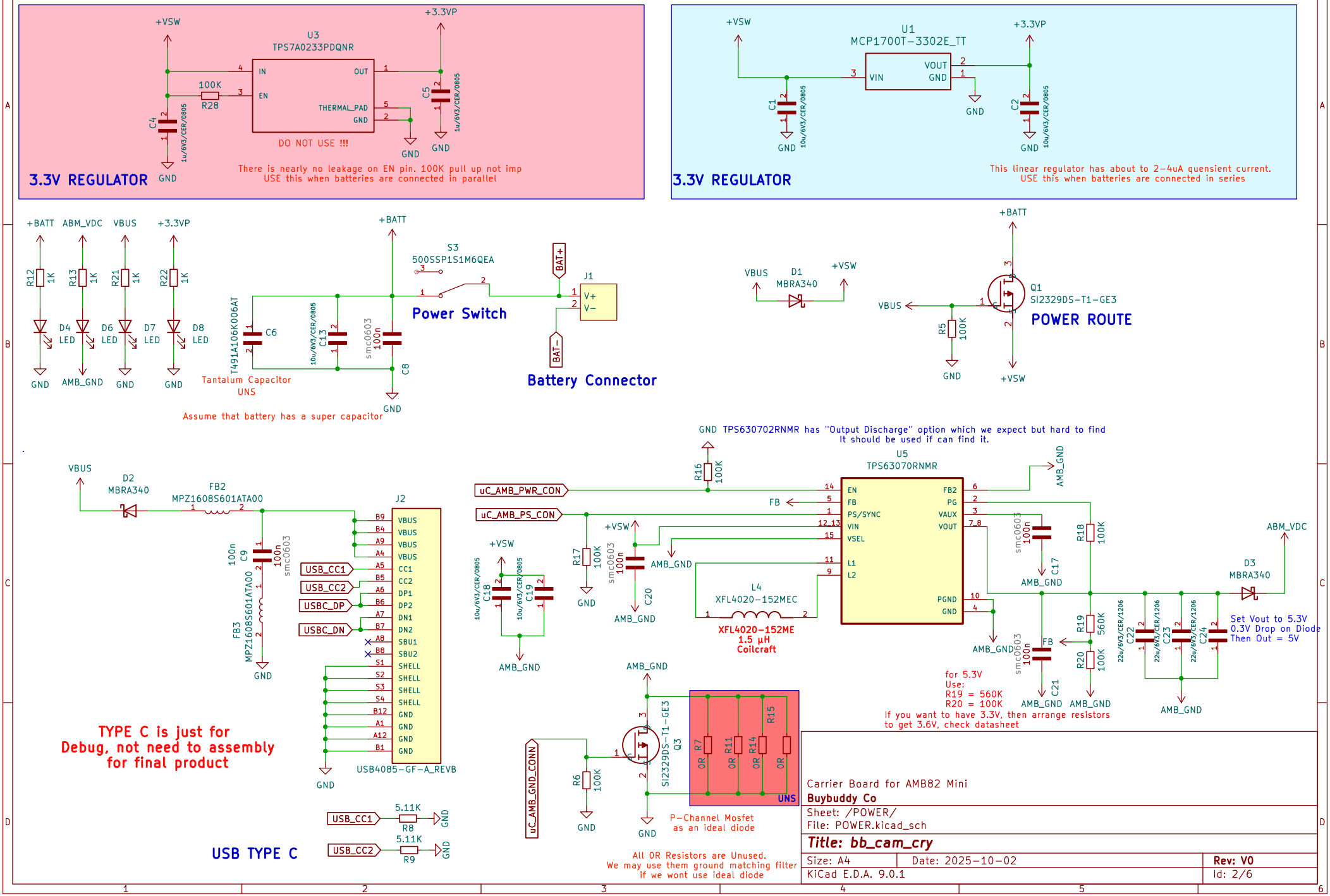
Buybuddy Co

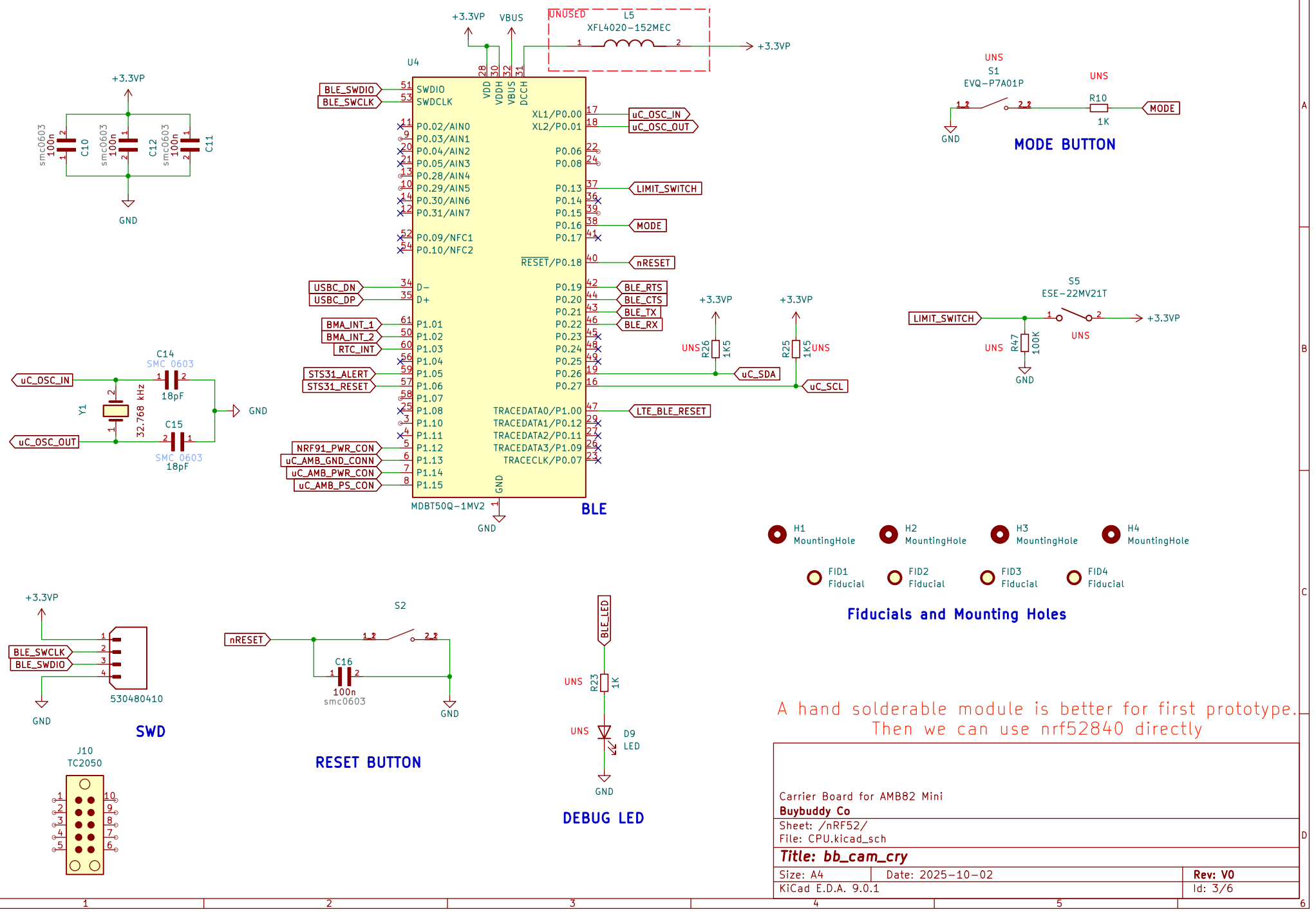
Sheet: /

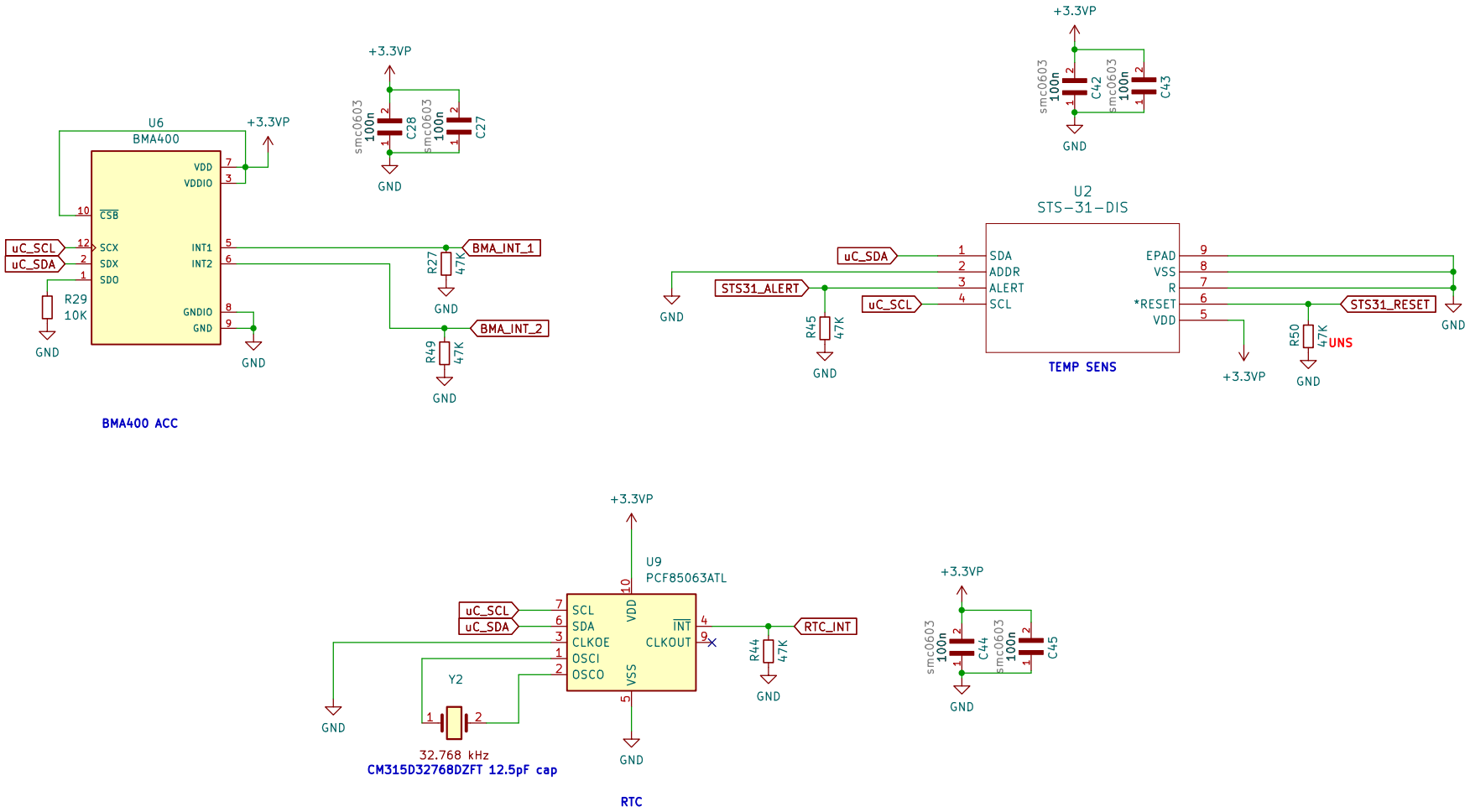
File: bb_cam_cry_v0.kicad_sch

Title: bb_cam_cry

Size: A4	Date: 2025-10-02	**Rev: V0**
KiCad E.D.A. 9.0.1		Id: 1/6


3.3V REGULATOR
U3
TPS7A0233PDQNR
+VSW
+3.3VP
100K
R28
C4
C5
DO NOT USE !!!
There is nearly no leakage on EN pin. 100K pull up not imp
USE this when batteries are connected in parallel
3.3V REGULATOR
U1
MCP1700T-3302E_TT
This linear regulator has about to 2-4uA quensient current.
USE this when batteries are connected in series
+BATT
ABM_VDC
VBUS
R12 1K
R13 1K
R21 1K
R22 1K
D4 LED
D6 LED
D7 LED
D8 LED
AMB_GND
S3
500SSP1S1M6QEA
Power Switch
J1
BAT+
BAT-
V+
V-
C6
T491A106K006AT
Tantalum Capacitor
UNS
C13
C8
Assume that battery has a super capacitor
Battery Connector
D1
MBRA340
Q1
SI2329DS-T1-GE3
POWER ROUTE
R5 100K
GND TPS630702RNMR has "Output Discharge" option which we expect but hard to find
It should be used if can find it.
U5
TPS63070RNMR
D2
FB2
MPZ1608S601ATA00
FB3
J2
USB_CC1
USB_CC2
USBC_DP
USBC_DN
USB4085-GF-A_REVB
TYPE C is just for
Debug, not need to assembly
for final product
USB TYPE C
5.11K
R8
R9
R16 100K
R17 100K
R6 100K
uC_AMB_PWR_CON
uC_AMB_PS_CON
uC_AMB_GND_CONN
L4
XFL4020-152MEC
XFL4020-152ME
1.5 µH
Coilcraft
Q3
SI2329DS-T1-GE3
R7
R11
R14
R15
OR
UNS
P-Channel Mosfet
as an ideal diode
All OR Resistors are Unused.
We may use them ground matching filter
if we wont use ideal diode
R18 100K
R19 560K
R20 100K
D3
MBRA340
Set Vout to 5.3V
0.3V Drop on Diode
Then Out = 5V
for 5.3V
Use:
R19 = 560K
R20 = 100K
If you want to have 3.3V, then arrange resistors
to get 3.6V, check datasheet
Carrier Board for AMB82 Mini
Buybuddy Co
Sheet: /POWER/
File: POWER.kicad_sch
Title: bb_cam_cry
Size: A4
Date: 2025-10-02
Rev: V0
KiCad E.D.A. 9.0.1
Id: 2/6


+3.3VP
VBUS
UNUSED
L5
XFL4020-152MEC
U4
BLE_SWDIO
BLE_SWCLK
SWDIO
SWDCLK
VDD
VDDH
VBUS
DCCH
XL1/P0.00
XL2/P0.01
uC_OSC_IN
uC_OSC_OUT
P0.02/AIN0
P0.03/AIN1
P0.04/AIN2
P0.05/AIN3
P0.28/AIN4
P0.29/AIN5
P0.30/AIN6
P0.31/AIN7
P0.06
P0.08
P0.13
P0.14
P0.15
P0.16
P0.17
LIMIT_SWITCH
MODE
P0.09/NFC1
P0.10/NFC2
RESET/P0.18
nRESET
USBC_DN
USBC_DP
D-
D+
P0.19
P0.20
P0.21
P0.22
P0.23
P0.24
P0.25
P0.26
P0.27
BLE_RTS
BLE_CTS
BLE_TX
BLE_RX
BMA_INT_1
BMA_INT_2
RTC_INT
P1.01
P1.02
P1.03
P1.04
P1.05
P1.06
P1.07
STS31_ALERT
STS31_RESET
P1.08
P1.10
P1.11
P1.12
P1.13
P1.14
P1.15
TRACEDATA0/P1.00
TRACEDATA1/P0.12
TRACEDATA2/P0.11
TRACEDATA3/P1.09
TRACECLK/P0.07
LTE_BLE_RESET
NRF91_PWR_CON
uC_AMB_GND_CONN
uC_AMB_PWR_CON
uC_AMB_PS_CON
GND
MDBT50Q-1MV2
BLE
UNS
S1
EVQ-P7A01P
R10
1K
MODE BUTTON
S5
ESE-22MV21T
LIMIT_SWITCH
R47
100K
R26
1K5
R25
1K5
uC_SDA
uC_SCL
C10
C11
C12
smc0603
100n
C14
SMC 0603
18pF
C15
Y1
32.768 kHz
uC_OSC_IN
uC_OSC_OUT
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
Fiducials and Mounting Holes
BLE_SWCLK
BLE_SWDIO
530480410
SWD
S2
nRESET
C16
100n
RESET BUTTON
BLE_LED
R23
1K
D9
LED
DEBUG LED
J10
TC2050
A hand solderable module is better for first prototype.
Then we can use nrf52840 directly
Carrier Board for AMB82 Mini
Buybuddy Co
Sheet: /nRF52/
File: CPU.kicad_sch
Title: bb_cam_cry
Size: A4
Date: 2025-10-02
Rev: V0
KiCad E.D.A. 9.0.1
Id: 3/6



Carrier Board for AMB82 Mini

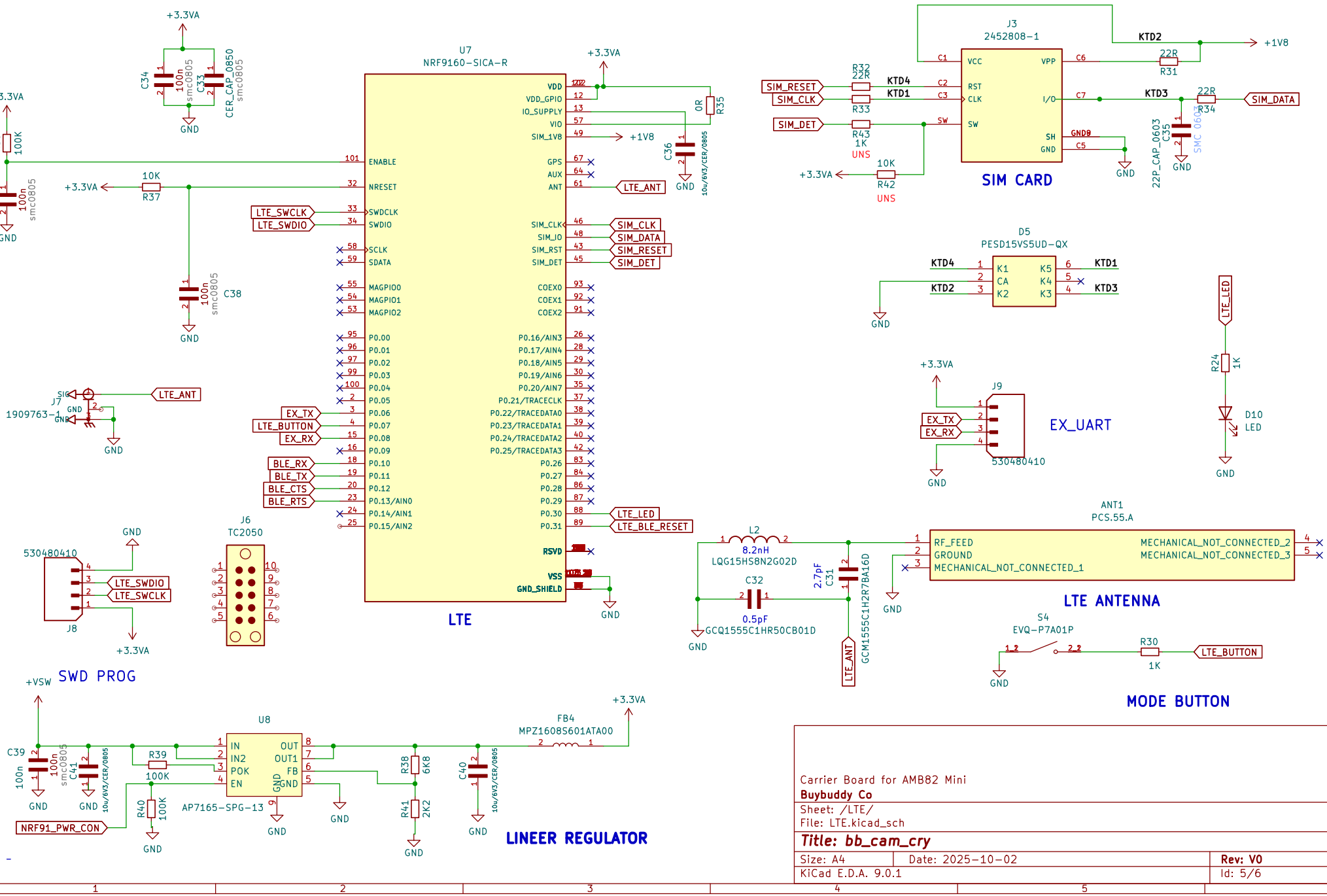
Buybuddy Co

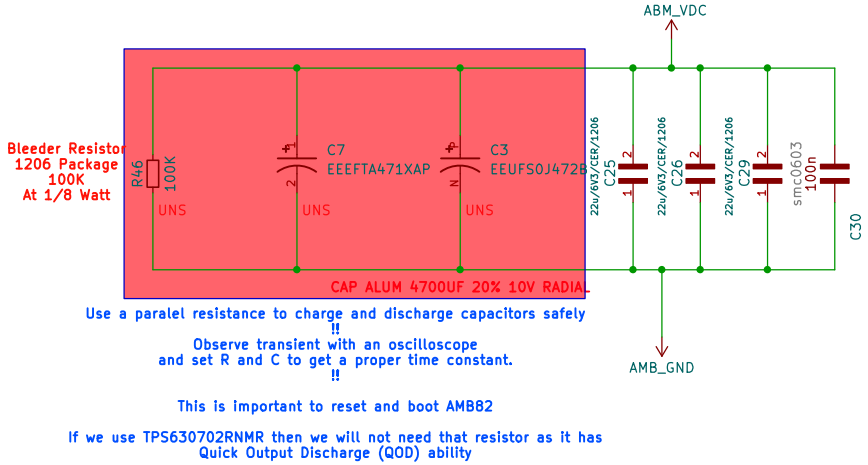
Sheet: /SENSORS/

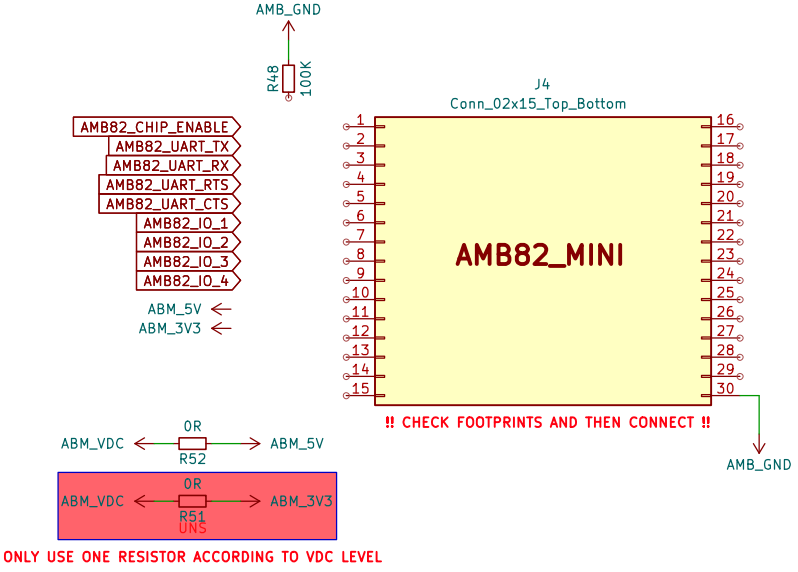
File: SENSOR.kicad_sch

Title: bb_cam_cry

Size: A4	Date: 2025-10-02	**Rev: V0**
KiCad E.D.A. 9.0.1		Id: 4/6


U7
NRF9160-SICA-R
LTE
SIM CARD
J3
2452808-1
D5
PESD15VS5UD-QX
EX_UART
J9
530480410
LTE ANTENNA
ANT1
PCS.55.A
MODE BUTTON
S4
EVQ-P7A01P
SWD PROG
J8
J6
TC2050
J7
1909763-1
U8
AP7165-SPG-13
LINEER REGULATOR
FB4
MPZ1608S601ATA00
L2
8.2nH
LQG15HS8N2G02D
C32
0.5pF
GCQ1555C1HR50CB01D
C31
2.7pF
GCM1555C1H2R7BA16D
D10
LED
Carrier Board for AMB82 Mini
Buybuddy Co
Sheet: /LTE/
File: LTE.kicad_sch
Title: bb_cam_cry
Size: A4
Date: 2025-10-02
Rev: V0
KiCad E.D.A. 9.0.1
Id: 5/6





Carrier Board for AMB82 Mini

Buybuddy Co

Sheet: /amb82_conn/

File: amb82_conn.kicad_sch

Title: bb_cam_cry

Size: A4	Date: 2025-10-02	**Rev: V0**
KiCad E.D.A. 9.0.1		Id: 6/6